Question	Answer	Marks	Guidance
1	**Method of Analysis**		Annotate with ✓
	Plot a graph of lg T against lg a **or** lg T against lg $\frac{1}{a}$ **or** ln T against ln a or equivalent	**1**	**Note** drawn axes take priority. **Allow** log for lg
	$q = -$gradient for lg T against lg a **or** $q =$ gradient for lg T against lg $\frac{1}{a}$	**1**	**Note** Must be consistent with drawn graph. $q = \frac{1}{\text{gradient}}$ for lg a against lg $\frac{1}{T}$
	$P = \frac{1}{2\pi}\sqrt{\frac{M}{L}} \times 10^{y\text{-intercept}}$ for lg T against lg a or lg T against lg $\frac{1}{a}$ **or** $P = \frac{1}{2\pi}\sqrt{\frac{M}{L}} \times e^{y\text{-intercept}}$ for ln T against ln a	**1**	**Note** Must be consistent with drawn graph. $P = \frac{1}{2\pi}\sqrt{\frac{M}{L}} \times 10^{q\times y\text{-intercept}}$ for lg a against lg $\frac{1}{T}$
	Additional detail including safety considerations	**6**	Annotate with numbered tick
	D1 Precaution linked to bar / rod falling onto bench, e.g. G-clamp stand(s) to bench to prevent bar / rod falling **or** cushions / sand tray (on bench / on the surface immediately below bar) to prevent bar / rod falling		**Allow** heavy weights on stand(s) **Ignore** cushions / sand tray on floor for use of bench **Note** position may be confirmed from diagram
	D2 Keep M constant		**Allow** same for constant
	D3 Method to measure M, e.g. use a (top-pan) balance		**Allow** scales **Ignore** scale **Allow** spring balance **Allow** newton meter / weight with division by g

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1	D4 Description of method to ensure each *a* is measured from the centre of the bar, e.g. measure length of bar and divide by two and mark midpoint of bar **or** measure from the end of the bar to each string to ensure each *a* is the same		**Note** text mark **Allow** determine balance point and mark the balance point
	D5 Description of method to keep bar horizontal e.g. use a spirit level or measure *L* at different positions on the bar / measure height of bar from the bench in different positions to check *L* / height are the same **or** description of method to keep the strings vertical, e.g. check with set square against bar or rod and string / measure *a* to each string at the bar and at the rod to check that they are same / use a plumb line		**Note** text mark **Allow** level for horizontal **Ignore** rod
	D6 Description of method to prevent strings from sliding, e.g. use adhesive putty to prevent the strings from sliding		**Allow** notches or holes or grooves
	D7 Method to determine accurate values of *T*, e.g. use small amplitude / angle oscillations or use a fiducial marker at the midpoint of the oscillation		**Allow** displacement **Ignore** not (too) large
	D8 Use large values of *L* to increase *T* (to reduce percentage uncertainty of *T*)		**Allow** *t* for *T*
	D9 Repeat determination of *T* for the same value of *a* **and** calculate the average *T*		**Allow** *t* for *T* **Allow** each for same

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Question	Answer	Marks	Guidance
1	D10 Relationship valid <u>if</u> a straight line is produced (with y-intercept = $\lg\left(2\pi P\sqrt{\frac{L}{M}}\right)$). Do not accept through the origin.		**Note** Must be consistent with drawn graph. **Ignore** 'data' / 'graph' / 'experiment' / 'results' / 'correlation' valid.

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